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9.0 INTRODUCTION

MANET stands for Mobile Adhoc Network also called a wireless Adhoc network or Adhoc wireless network that usually has a routable networking environment on top of a Link Layer ad hoc network.. They consist of a set of mobile nodes connected wirelessly in a self-configured, self-healing network without having a fixed infrastructure. MANET nodes are free to move randomly as the network topology changes frequently. Each node behaves as a router as they forward traffic to other specified nodes in the network.

Mobile Ad Hoc Networks (MANETs) are wireless communication networks composed of mobile devices that can dynamically form temporary connections without the need for a fixed infrastructure. In a MANET, devices, often referred to as nodes, communicate directly with each other, creating a self-organizing network that can adapt to changing conditions and scenarios.

Designing MANETs involves creating efficient routing protocols, ensuring security mechanisms, managing energy consumption, and addressing the unique challenges posed by node mobility and dynamic topology changes.

As technology advances, MANETs continue to find applications in diverse fields and contribute to the development of innovative networking solutions that adapt to the demands of dynamic and unpredictable environments.

9.1 OBJECTIVES

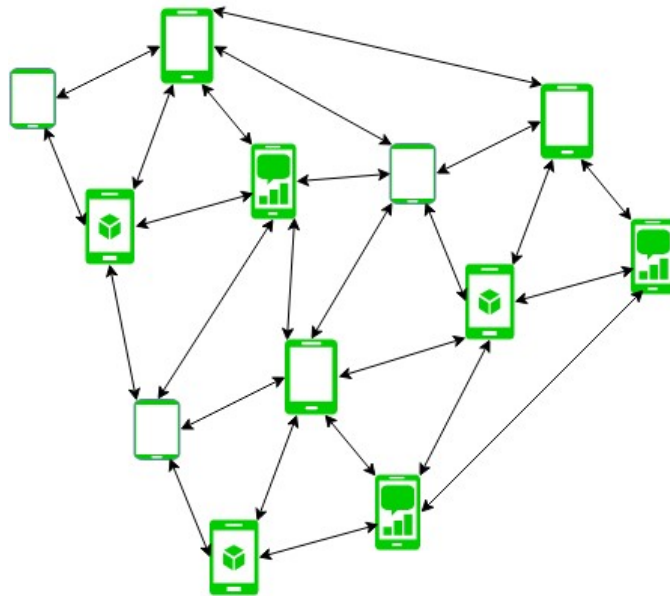
After going through this unit, you should be able to learn,

Mobile Ad Hoc Networks (MANETs) have several key objectives that drive their design, deployment, and use. These objectives are shaped by the unique characteristics of MANETs and the specific challenges they aim to address.

Overall, the objectives of MANETs revolve around creating self-organizing, decentralized networks that can operate in challenging and dynamic environments. These networks cater to a wide range of applications and provide solutions where traditional communication methods may not be feasible or efficient.

9.2 INTRODUCTION TO MANETs

A Mobile Ad hoc Network (MANET) is a decentralized wireless network where devices communicate directly with each other without relying on a fixed infrastructure like routers or access points. Devices in a MANET form temporary connections, dynamically creating a network on-the-fly as shown in 9.2. Figure: Mobile Ad Hoc Network. These networks are often used in situations where traditional infrastructure is unavailable such as disaster recovery or military operations.



9.2 Figure: Mobile Ad Hoc Network

MANET may operate a standalone fashion or they can be part of larger internet. They form a highly dynamic autonomous topology with the presence of one or multiple different transceivers between nodes. The main challenge for the MANET is to equip each device to continuously maintain the information required to properly route traffic. MANETs consist of a peer-to-peer, self-forming, self-healing network.

MANETs are characterized by their unique features:

- **Decentralization:** Unlike traditional networks, MANETs lack a central control point. Each node acts as both a user and a router, forwarding data for other nodes.

- **Dynamic Topology:** The topology of a MANET changes frequently due to node mobility. Nodes can join or leave the network, and connections between nodes can be established or broken at any time.
- **Resource Constraints:** Devices in a MANET may have limited processing power, memory, battery life, and wireless communication range, which necessitate the development of energy-efficient protocols and algorithms.
- **Ad Hoc Formation:** MANETs can be deployed quickly without the need for a pre-existing infrastructure. This makes them suitable for situations where traditional networks are unavailable, such as disaster-stricken areas or battlefields.
- **Challenges:** MANETs face challenges like routing in dynamic environments, ensuring security against various threats, maintaining Quality of Service (QoS) for different applications, and managing energy consumption efficiently.

9.2.1 Applications of MANETs

Some of the main applications of MANETs including are:

1. **Dynamic Communication:** MANETs aim to establish communication links among mobile devices without relying on fixed infrastructure. This allows for flexible and dynamic network formation in scenarios where traditional infrastructure is unavailable or impractical.
2. **Decentralization:** MANETs are designed to operate without a centralized control point. Each node in the network acts as both a user and a router, contributing to the overall network functionality. This decentralization enhances network resilience and adaptability.
3. **Flexibility and Scalability:** MANETs can be quickly deployed and scaled up or down based on the number of participating devices. This flexibility is advantageous in situations where network setup needs to be rapid and adaptable.
4. **Robustness and Resilience:** MANETs are built to tolerate node failures, mobility, and changing network topologies. They can maintain connectivity and route data even in dynamic environments, making them suitable for scenarios such as disaster recovery and battlefield communications.
5. **Emergency and Remote Communication:** MANETs are often used in emergency situations where traditional communication infrastructure is unavailable or damaged. They enable first responders to establish communication quickly and effectively in disaster-stricken areas.
6. **Military Applications:** In military operations, MANETs allow for secure and dynamic communication among soldiers, vehicles, and other assets. Their self-configuring nature makes them suitable for tactical environments.
7. **Sensor Networks:** MANETs are used in sensor networks for applications like environmental monitoring, wildlife tracking, and industrial automation. These networks can be deployed in remote or hostile areas where traditional communication is challenging.
8. **Collaborative Applications:** MANETs support collaborative applications, such as sharing resources, data, and information among devices in close

proximity. This is useful for scenarios like interactive presentations or collaborative workgroups.

9. **Reduced Infrastructure Costs:** By eliminating the need for fixed infrastructure, MANETs can reduce the costs associated with deploying and maintaining traditional communication infrastructure.
10. **Research and Development:** MANETs serve as a platform for research in various fields, including routing protocols, energy-efficient communication, security mechanisms, and adaptive networking strategies.
11. **Community Networks:** In rural or underserved areas, MANETs can be established to provide local connectivity and access to information, bridging the digital divide.

Applications of MANETs include:

Disaster Recovery: MANETs enable communication in disaster-stricken areas where traditional infrastructure is damaged or unavailable.

Military Operations: They support secure communication between soldiers, vehicles, and bases in battlefield scenarios.

Emergency Response: First responders can establish communication quickly in emergency situations, improving coordination and aid distribution.

Sensor Networks: MANETs are used in sensor deployments for environmental monitoring, wildlife tracking, and industrial automation.

Collaborative Work: They facilitate communication in interactive presentations, collaborative workgroups, and entertainment scenarios.

Remote Areas: MANETs can provide connectivity in rural or underserved regions, helping bridge the digital divide.

Check Your Progress 1

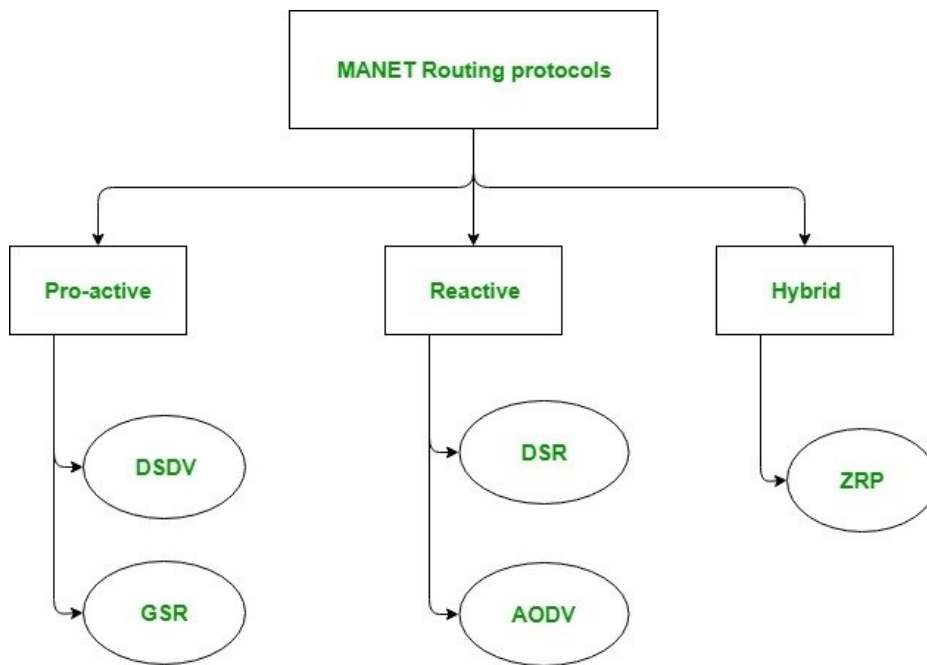
1. MANETs allow for-----among soldiers, vehicles, and other assets.
2. MANETs support-----such as sharing resources, data, and information among devices in close proximity.
3. What are the Applications of MANETs?

9.3 ROUTING AND CLASSIFICATION OF ROUTING ALGORITHMS

The Essential Information of Routing and Classifications of Its Algorithms

Routing refers to the process of determining the best path for data packets to travel from their source to destination in a network. It involves in making decisions based on network protocols and algorithms to efficiently direct data through various network devices such as routers, switches, and gateways. The goal of routing is to ensure that data packets reach their intended destination reliably and in a timely manner.

The choice of routing algorithm depends on factors such as the network size, complexity, scalability, reliability requirements, and the nature of the data being transmitted.



9.3. Figure: Routing Algorithm Classification

The above 9.3.Figure shows, Routing Algorithms Classification in mobile ad hoc networks (MANETs) determine how data packets are forwarded from one device to another within the network. They can be classified into several categories based on their approach and characteristics:

9.3.1 Proactive (Table-Driven) Routing: These are also known as table-driven routing protocols. Each mobile node maintains a separate routing table which contains the information of the routes to all the possible destination mobile nodes. Since the topology in the mobile ad-hoc network is dynamic, these routing tables are updated periodically as and when the network topology changes. It has a limitation that it doesn't work well for the large networks as the entries in the routing table becomes too large since they need to maintain the route information to all possible nodes.

These algorithms maintain up-to-date routing tables for all nodes in the network. Examples include Optimized Link State Routing (OLSR) and Destination-Sequenced Distance Vector (DSDV). They work well for relatively stable networks but can incur high overhead due to continuous table updates.

TABLE-DRIVEN ROUTING PROTOCOLS

These protocols are extensions of the wired network routing protocols, they maintain global topology information in the form of tables at every node. Table are updated frequently in order to maintain consistent and accurate network state information.

Ex: Destination Sequenced Distance Vector routing protocol (DSDV), wireless routing protocol (WRP), source-tree adaptive routing protocol (STAR) and cluster-head gateway switch routing protocol (CGSR).

9.3.1.1 Destination Sequenced Distance-Vector (DSDV) Routing Protocol

It is an enhanced version of the distributed Bellman -Ford algorithm where each node maintains a table that contains the shortest distance and the first node on the shortest path to every other node in the network.

It incorporates table updates with increasing sequence number tags to prevent loops, to counter the count-to-infinity problem, and for faster convergence.

As it is a table-driven routing protocol, routes to all destinations are readily available at every node at all times.

The tables are exchanged between neighbours at regular intervals to keep an up -to-date view of the network topology.

The table updates are of two types:

Incremental updates: Takes a single network data packet unit (NDPU). These are used when a node does not observe significant changes in the local topology.

Full dumps: Takes multiple NDPUs. It is done either when the local topology changes significantly or when an incremental update requires more than a single NDPU.

Table updates are initiated by a destination with a new sequence number which is always greater than the previous one.

Consider the example as shown in figure 9.3.1.1 (a). Here node 1 is the source node and node 15 is the destination. As all the nodes maintain global topology information, the route is already available as shown in figure 9.3.1.1 (b).

Here the routing table node 1 indicates that the shortest route to the destination node is available through node 5 and the distance to it is 4 hops, as depicted in figure (b)

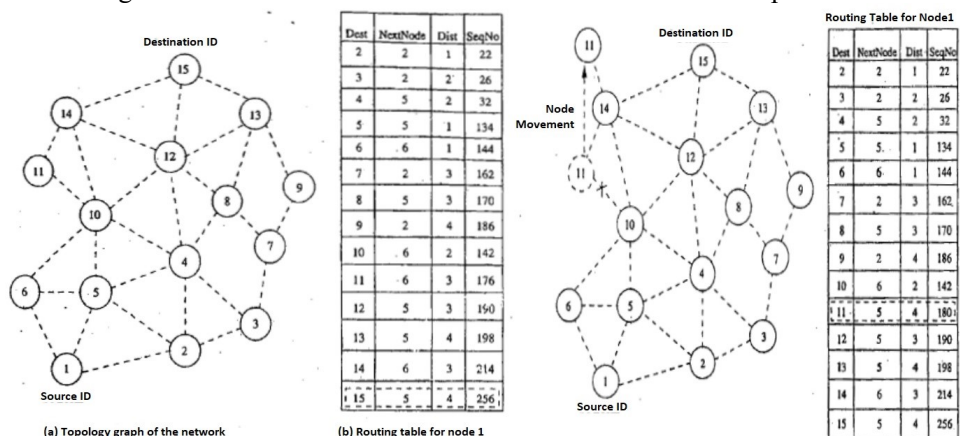
The reconfiguration of a path used by an on-going data transfer session is handled by the protocol in the following way.

The end node of the broken link initiates a table update message with the broken link's weight assigned to infinity (∞) and with a sequence number greater than the stored sequence number for that destination.

Each node upon receiving an update with weight ∞ , quickly disseminates it to its neighbours in order to propagate the broken-link information to the whole network.

A node always assign an odd number to the link break update to differentiate it from the even sequence number generated by the destination.

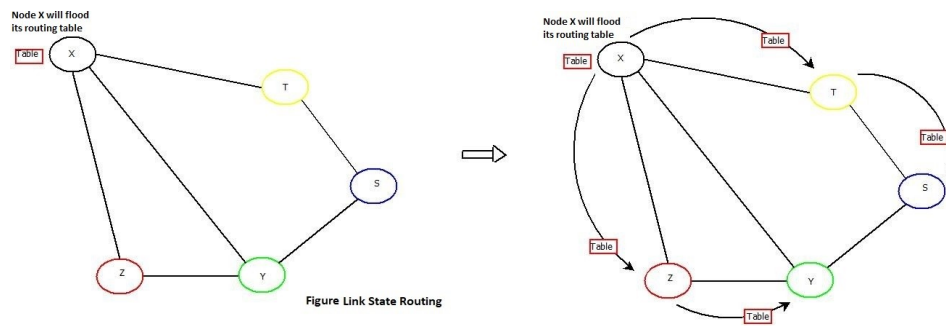
9.3.1.1. Figure: shows the case when node 11 moves from its current position.



9.3.1.1. Figure : Route Establishment in DSDV

9.3.1.1. Figure: Route Maintenance in DSDV

- Global State Routing is based upon the fundamental concepts of link state routing.
- In Link State Routing (LSR) shows in 9.3.1.2 Figure, one of the node floods out single routing table information to its neighbours and those neighbours' floods out that table to further nodes. This process continues to take place until the routing table is received by all the nodes throughout the network.
- But in case of Global State Routing, the routing table of a particular node is broadcasted to its immediate neighbours only. The initial tables of those neighbouring nodes are updated. These updated tables are further broadcast one by one and this process continues to take place until all nodes broadcasts their tables to each node in the network.



9.3.1.2 Figure: Link State Routing

Basic Concept of Global State Routing is,

GSR protocol uses and maintains three tables for every node individually. These tables are:

1. **Distance Table:** This table contains the distance of a node from all the nodes in network.

Format :

Node	Distance
------	----------

Global State Routing: Distance Table

2. **Topology Table:** This table contains the information of Link state data along with the sequence number which can be used to determine when the information is updated last.

Format :

Node	Link State	Sequence
------	------------	----------

Global State Routing: Topology Table

3. **Next Hop Table:** Next hop table will contain the information about the immediate neighbour of a particular node.

Format :

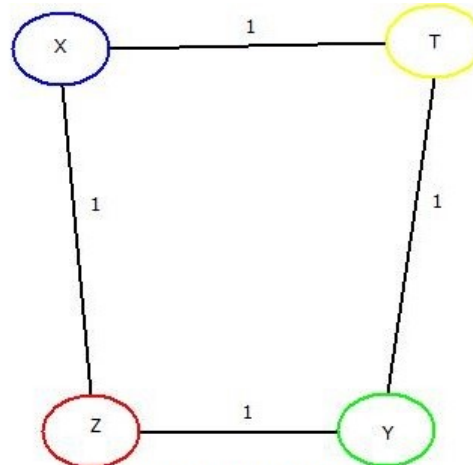
Node	Next Hop
------	----------

Global State Routing: Next Hop Table

These tables are updated on every step and ensure that each node receives correct information about all the nodes including their distances.

Global State Routing Protocol: Working

- GSR broadcasts the routing tables to its immediate neighbours rather than flooding it to all the nodes as Link State Routing protocol does.
- Consider a network of 4 nodes having a distance of “1” on each of its edge in below figure 9.3.1.2.1 GSRP:W. Below mentioned steps will let you know how GSR works and how its routing tables are updated



9.3.1.2.1. Figure:GSRP.Working

Steps

1. For Node “X”: Firstly three tables as mentioned above will be maintained which includes distance table, Topology table and Next hop tables. This same process will be done for rest of the nodes too.

For X:

Topology Table			Next Hop Table		Distance Table	
Node	Link State	Sequence	Node	Link State	Node	Distance
X	{}	----	X	X	X	0
Y	{}	----	Y	-1	Y	Infinite
Z	{}	----	Z	-1	Z	Infinite
T	{}	----	T	-1	T	Infinite

2. Secondly, broadcasting of all the tables will be done to all the immediate neighbors of “X” i.e. “Y” and “Z”.
3. These tables are updated at “X”, “Y” & “T” nodes respectively.
4. Same will be done for node “Y”. After first updation from “X”, node “Y” will broadcast the tables to its immediate neighbors i.e. “X” & “T” and those tables will be updated accordingly. This will be done for “T” & “Z” also.
5. Once done, all the nodes “X”, “Y”, “Z” & “T” will be having the updated routing tables containing distances from each, with the help of which an optimal path can be chosen if data needs to be transferred from one node to other.

For X:

Distance Table		Next Hop Table		Topology Table		
Node	Distance	Node	Next	Node	Link State	SEQ Number
X	0	X	X	X	{Y,Z}	1
Y	1	Y	Y	Y	{}	----
Z	1	Z	Z	Z	{X,T}	----
T	∞	T	-1	T	{}	----

For Y:

Distance Table		Next Hop Table		Topology Table		
Node	Distance	Node	Next	Node	Link State	SEQ Number
X	1	X	X	X	{}	----
Y	0	Y	Y	Y	{X,T}	1
Z	∞	Z	-1	Z	{}	----
T	1	T	T	T	{}	----

For Z:

Distance Table		Next Hop Table		Topology Table		
Node	Distance	Node	Next	Node	Link State	SEQ Number
X	1	X	X	X	{}	----
Y	∞	Y	-1	Y	{}	----
Z	0	Z	-Z	Z	{X,T}	1
T	1	T	T	T	{}	----

6. Now, broadcasting of topology tables of "X" will take place to its neighbor i.e. "Y" & "Z" and updated tables will be like as mentioned below.

For Y:

Distance Table		Next Hop Table		Topology Table		
Node	Distance	Node	Next	Node	Link State	SEQ Number
X	1	X	X	X	{Y,Z}	1
Y	0	Y	Y	Y	{X,T}	1
Z	2	Z	X	Z	{}	----
T	1	T	T	T	{}	----

For Z:

Distance Table		Next Hop Table		Topology Table		
Node	Distance	Node	Next	Node	Link State	SEQ Number
X	1	X	X	X	{Y,Z}	1
Y	2	Y	X	Y	{}	----
Z	0	Z	Z	Z	{X,T}	1
T	1	T	T	T	{}	----

7. Similarly, these tables are further updated with topology tables of "Y", "Z" & "T" as done in case of "X".

Advantages : Global State Routing Protocol

- Higher accuracy of GSR in generating optimal path as compared to LSR.
- Broadcasting reduces error rate as compare to flooding used in LSR.

Disadvantages : Global State Routing Protocol

- Large bandwidth consumption.
- Higher operational cost.
- Large Message size resulting in more time consumption.

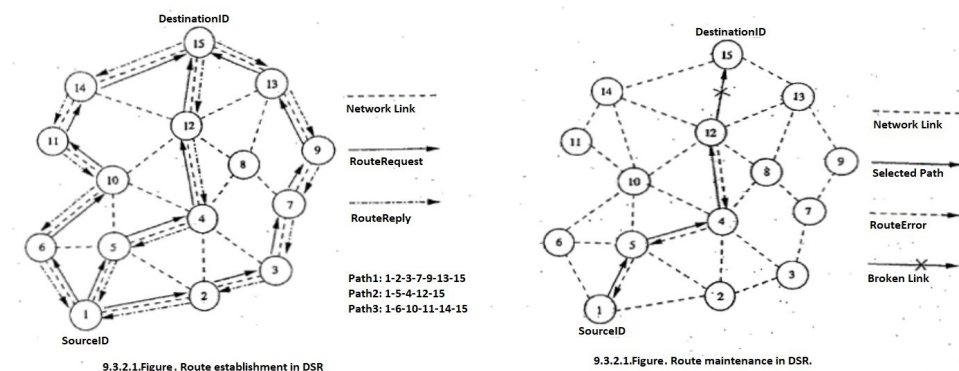
9.3.2. Reactive (On-Demand) Routing:

These are also known as on-demand routing protocol. In this type of routing, the route is discovered only when it is required/needed. The process of route discovery occurs by flooding the route request packets throughout the mobile network. It consists of two major phases namely, route discovery and route maintenance.

9.3.2.1 Dynamic Source Routing protocol (DSR): It is a reactive/on-demand routing protocol. In this type of routing, the route is discovered only when it is required/needed. The process of route discovery occurs by flooding the route request packets throughout the mobile network. In this protocol, Source node stores the complete path information and intermediate nodes do not need to maintain routing information. It consists of two phases:

Route Discovery: This phase determines the most optimal path for the transmission of data packets between the source and the destination mobile nodes.

Route Maintenance: This phase performs the maintenance work of the route as the topology in the mobile ad-hoc network is dynamic in nature and hence, there are many cases of link breakage resulting in the network failure between the mobile nodes.



Designed to restrict the bandwidth consumed by control packets in adhoc wireless networks by eliminating the periodic table update messages. It is beacon-less and does not require periodic hello packet transmissions.

Basic approach → to establish a route by flooding RouteRequest packets in the network. Destination node responds by sending a RouteReply packet back to the source.

Each RouteRequest carries a sequence number generated by the source node and the path it has traversed. A node checks the sequence number on the packet before forwarding it. The packet is forwarded only if it is not a duplicate RouteRequest.

The sequence number on the packet is used to prevent loop formations and to avoid multiple transmissions.

Thus, all nodes except the destination forward a RouteRequest packet during the route construction phase. In figure 9.3.2.1, source node 1 initiates a RouteRequest packet to obtain a path for destination node 15. This protocol uses a route cache that stores all possible information extracted from the source route contained in a data packet. During network partitions, the affected nodes initiate RouteRequest packets.

DSR also allows piggy-backing of a data packet on the RouteRequest. As a part of optimizations, if the intermediate nodes are also allowed to originate RouteReply packets, then a source node may receive multiple replies from intermediate nodes.

In figure 9.3.2.1, if the intermediate node 10 has a route to the destination via node 14, it also sends the RouteReply to the source node. The source node selects the latest and best route and uses that for sending data packets. Each data packet carries the complete path to its destination. If a link breaks, source node again initiates the route discovery process.

Advantages

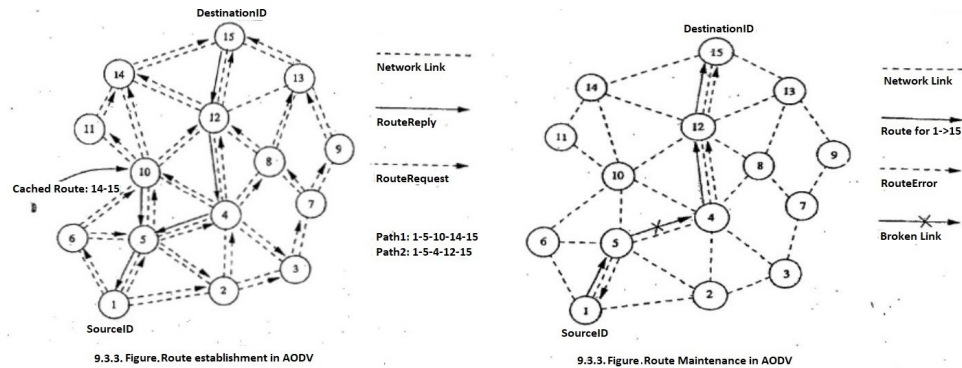
- Uses a reactive approach which eliminates the need to periodically flood the network with table update messages.
- Route is established only when required.
- Reduce control overhead.

Disadvantages

- Route maintenance mechanism does not locally repair a broken link.
- Stale route cache information could result in inconsistencies during route construction phase.
- Connection set up delay is higher.
- Performance degrades rapidly with increasing mobility.
- Routing overhead is more & directly proportional to path length.

9.3.3 Ad-Hoc On Demand Vector Routing protocol (AODV): It is a reactive/on-demand routing protocol. It is an extension of dynamic source routing protocol (DSR) and it helps to remove the disadvantage of dynamic source routing protocol. In DSR, after route discovery, when the source mobile node sends the data packet to the destination mobile node, it also contains the complete path in its header. Hence, as the network size increases, the length of the complete path also increases and the data packet's header size also increases which makes the whole network slow.

Hence, Ad-Hoc On Demand Vector Routing protocol came as solution to it. The main difference lies in the way of storing the path, in AODV Source node does not store complete path information, instead of that each node stores information of its previous and next node. As shown in figure 9.3.3, it also operates in two phases: Route discovery and Route maintenance.



Reactive algorithms establish routes only when needed, reducing overhead. When a device wants to send data, it initiates route discovery. Examples include Ad Hoc On-Demand Distance Vector (AODV) and Dynamic Source Routing (DSR).

9.3.3. Hybrid Routing

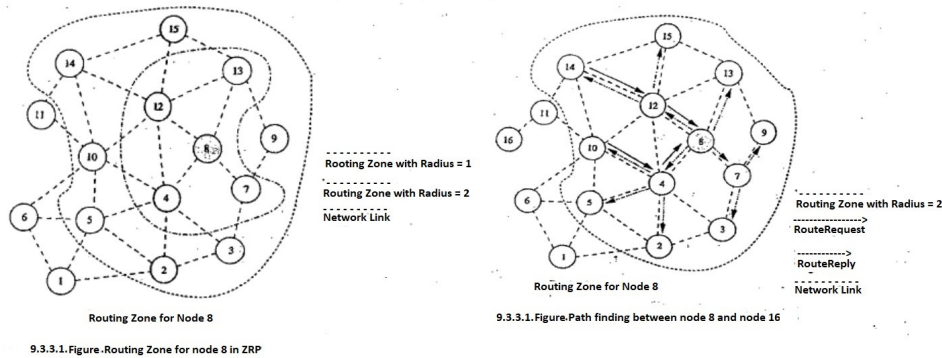
It basically combines the advantages of both, reactive and pro-active routing protocols. These protocols are adaptive in nature and adapts according to the zone and position of the source and destination mobile nodes. One of the most popular hybrids routing protocol is **Zone Routing Protocol (ZRP)**.

The whole network is divided into different zones and then the position of source and destination mobile node is observed. If the source and destination mobile nodes are present in the same zone, then proactive routing is used for the transmission of the data packets between them. And if the source and destination mobile nodes are present in different zones, then reactive routing is used for the transmission of the data packets between them.

These algorithms combine elements of both proactive and reactive approaches. They maintain routing information for some nodes while using on-demand routing for others. Zone Routing Protocol (ZRP) is an example of a hybrid algorithm.

9.3.3.1 Zone Routing Protocol (ZRP)

Effectively combines the best features of both Proactive and Reactive routing protocols. It use a Proactive routing scheme within a limited zone in the r-hop neighbourhood of every node. Use a Reactive routing scheme for nodes beyond this. An Intra-Zone Routing Protocol (IARP) is used in the zone where a particular node employs proactive routing. The Reactive routing protocol used beyond this zone is referred to as Inter-Zone Routing Protocol (IERP). The routing zone of a given node is a subset of the network, within which all nodes are reachable within less than or equal to.



Route Establishment: When a node s (node 8 in the figure 9.3.3.1) has packets to be sent to a destination node d (node 15 in figure 9.3.3.1), it checks whether node d is within its zone. If the destination belongs to its own zone, then it delivers the packets directly.

Otherwise, node s broadcasts the RouteRequest to its peripheral nodes (in figure 9.3.3.1, node 8 broadcasts RouteRequest to node 2, 3, 5, 7, 9, 10, 13, 14 and 15).

If any peripheral node finds node d to be located within its routing zone, it sends a RouteReply back to node 8 indicating the path; otherwise, the node reorders casts the RouteRequest packet to the peripheral nodes.

This process continues until node d is located.

During RouteRequest propagation, every node that forwards the RouteRequest appends its address to it. This information is used for delivering the RouteReply packet back to the source.

The criteria for selecting the best path may be the shortest path, least delay path etc.

When an intermediate node in an active path detects a broken link in the path, it performs a local path reconfiguration in which the broken link is bypassed by means of a short alternate path connecting the ends of the broken link.

A path update message is then sent to the sender node, this results in sub-optimal path between two end points.

Advantage

Reduce the control overhead by combining the best features of Proactive and Reactive protocols.

Disadvantage

Control overhead may increase due to the large overlapping of nodes routing zones.

9.3.4. Geographic Routing:

These algorithms use location information of nodes to forward packets. Greedy Perimeter Stateless Routing (GPSR) is one such example, where packets are forwarded to the next closest node towards the destination.

Geographic routing is a routing paradigm used in wireless and mobile ad hoc networks, where routing decisions are based on the geographic locations of nodes rather than traditional identifiers like IP addresses. In geographic routing, nodes are aware of their own positions and the positions of their neighbors.

When a node wants to send a packet to a specific destination, it uses location information to determine the next hop towards that destination. The packet is forwarded to the neighbor that is geographically closest to the destination. This process continues until the packet reaches the intended destination.

Geographic routing can be efficient in scenarios where nodes have accurate location information and the network topology changes frequently, such as in mobile ad hoc networks or wireless sensor networks. However, challenges like holes in coverage, uneven node distribution, and the need for location services can impact the effectiveness of geographic routing.

9.3.5. QoS-based Routing:

Quality of Service (QoS) routing considers factors like bandwidth, latency, and reliability when selecting routes. These algorithms aim to meet specific performance requirements for different types of traffic.

Quality of Service (QoS)-based routing is a network routing approach that takes into consideration the quality of service requirements for data transmission, in addition to finding a feasible path from source to destination. QoS parameters can include factors like bandwidth, delay, jitter, packet loss, and reliability.

QoS-based routing aims to ensure that specific QoS requirements are met while routing data packets through the network. Different applications and services have varying QoS demands; for instance, real-time communication like VoIP or video streaming may require low latency and minimal packet loss, while file transfers may prioritize bandwidth efficiency.

QoS-based routing protocols evaluate various paths based on their ability to satisfy the QoS parameters specified by the application. This evaluation might involve considering network conditions, link capacities, and potential congestion points. If no path can meet the desired QoS requirements, the routing protocol might reject the transmission or choose a path that offers the best compromise.

This routing approach is crucial in ensuring that different types of traffic receive the appropriate treatment in networks, especially in scenarios where there are limited resources or varying network conditions.

9.3.6. Multipath Routing:

Multipath routing algorithms use multiple routes simultaneously to increase reliability and load balancing. They can be proactive or reactive, depending on the approach used to discover and maintain multiple paths.

Multipath routing is a networking approach where data is split and transmitted across multiple paths simultaneously to reach the destination. This technique aims to improve network efficiency, increase reliability, and balance load among different routes.

By using multiple paths, multipath routing can achieve several benefits:

1. **Load Balancing:** Distributing traffic across multiple paths helps prevent network congestion on a single route, leading to better utilization of available bandwidth.
2. **Redundancy:** If one path fails or experiences issues, the data can be rerouted through alternative paths, enhancing network reliability.
3. **Increased Throughput:** Multipath routing can potentially increase the overall throughput of data transmission by using multiple links concurrently.

4. **Reduced Latency:** By selecting less congested paths, multipath routing can help reduce end-to-end latency for data transmission.

However, multipath routing also comes with challenges, such as managing packet ordering, avoiding loops, and handling path changes dynamically. Routing protocols that support multipath routing need to be designed carefully to ensure proper functioning and optimal performance in complex network environments.

9.3.7. Power-Aware Routing:

These algorithms take into account the energy constraints of mobile devices and try to optimize routing paths to conserve battery power. They help prolong the network's overall lifetime.

Power-aware routing is a networking approach that takes into consideration the energy consumption of nodes in the network when making routing decisions. The goal of power-aware routing is to prolong the lifespan of energy-constrained devices, such as nodes in wireless sensor networks or mobile devices, by efficiently managing their energy resources.

Power-aware routing strategies aim to minimize the energy consumption of nodes while ensuring effective data delivery. This can involve:

Energy-Efficient Paths: Routing decisions consider the energy levels of nodes and prioritize paths that involve nodes with more energy to avoid early depletion of nodes with lower energy reserves.

Sleep Modes: Nodes might be put into sleep or low-power modes when not actively participating in data forwarding to conserve energy.

Adaptive Routing: Routing decisions can dynamically adapt based on changing energy levels of nodes or the network conditions.

Energy-Harvesting Nodes: Nodes that can harvest energy from the environment (e.g., solar panels) might be used strategically to extend network operation.

Power-aware routing is particularly important in scenarios where battery-powered devices need to operate for extended periods without frequent battery replacements or recharging. It's commonly used in applications like wireless sensor networks, Internet of Things (IoT) deployments, and mobile ad hoc networks.

9.3.8. Security-Aware Routing:

Security-aware algorithms prioritize secure communication and take measures to protect data transmission from various threats such as eavesdropping and malicious nodes.

These classifications provide a framework for understanding the diverse range of routing algorithms available for mobile ad hoc networks, each tailored to address different challenges and requirements within these dynamic and often resource-constrained environments.

Security-aware routing is a networking concept that considers security concerns when making routing decisions in a network. The primary goal of security-aware routing is to ensure that data is transmitted through paths that minimize the risk of data breaches, unauthorized access, and other security threats.

In security-aware routing:

Secure Paths: Routing decisions take into account the security features of available paths, such as encryption, authentication, and integrity checks.

Avoiding Vulnerable Nodes: Paths that involve nodes with known vulnerabilities or compromised security are avoided.

Dynamically Adjusting Routes: Routing decisions might change in response to security incidents or changes in network conditions to mitigate potential threats.

Attack Detection and Response: Security-aware routing protocols might incorporate mechanisms to detect and respond to potential attacks, rerouting traffic to avoid compromised areas.

Security-aware routing is particularly important in scenarios where data privacy and network integrity are crucial, such as in military networks, critical infrastructure systems, and sensitive data communication. It helps protect against eavesdropping, data manipulation, and other cyber threats that can undermine the confidentiality and reliability of network communication.

Check Your Progress 2

1. The goal of routing is to ensure that ----- reach their intended destination reliably and in a timely manner.
2. -----phase determines the most optimal path for the transmission of data packets between the source and the destination mobile nodes.
3. How to avoid vulnerable nodes?

9.4 QoS in MANETs

Quality of Service (QoS) in Mobile Ad Hoc Networks (MANETs) refers to the ability of the network to provide specific levels of performance for different types of traffic or applications. Due to the dynamic and resource-constrained nature of MANETs, ensuring consistent QoS can be challenging. Here are some key aspects of QoS in MANETs:

Performance Metrics: QoS in MANETs can be measured using various performance metrics such as throughput, delay, packet loss, jitter (variation in delay), and reliability. These metrics reflect the network's ability to meet the requirements of different applications.

Traffic Differentiation: MANETs often handle diverse types of traffic, including voice, video, and data. QoS mechanisms aim to prioritize and allocate resources based on the specific requirements of each traffic type. For example, real-time applications like voice and video may require low delay and jitter, while data applications might prioritize high throughput.

Routing and QoS: Some routing algorithms in MANETs take QoS into account when selecting routes. These algorithms consider factors such as available bandwidth, latency, and reliability to choose routes that best meet the QoS requirements of the traffic.

Resource Management: Effective resource management is crucial for maintaining QoS in MANETs. This includes managing bandwidth, power, and network capacity to

ensure that the network can accommodate various types of traffic without degradation in performance.

Congestion Control: Congestion can significantly degrade QoS. Congestion-aware algorithms detect and alleviate congestion to prevent network performance degradation. These mechanisms might involve rate limiting, flow control, and dynamic route adjustments.

Adaptive QoS: Due to the dynamic nature of MANETs, QoS mechanisms need to be adaptive. They should be able to react to changes in network topology, link conditions, and traffic load. Adaptive QoS techniques dynamically adjust resource allocation to maintain desired performance levels.

Cross-Layer Approaches: QoS can benefit from cross-layer interactions, where information from different layers of the protocol stack is used to make decisions. For instance, link quality information from the physical layer can be used by higher layers to make routing and scheduling decisions.

Challenges: MANETs face challenges such as limited bandwidth, node mobility, energy constraints, and network partitioning. These challenges can impact the ability to maintain consistent QoS levels.

Overall, QoS in MANETs involves a combination of routing protocols, resource management, and adaptive mechanisms to ensure that the network can meet the diverse requirements of various applications while adapting to the dynamic nature of the network environment.

Check Your Progress 3

1. -----algorithms detect and alleviate congestion to prevent network performance degradation.
2. Performance Metrics in QoS can be measured using -----
3. What are the challenges in MANETs?

9.5 SECURITY IN MANETs

Security is a critical concern in Mobile Ad Hoc Networks (MANETs) due to their decentralized, dynamic, and often open nature. The absence of a fixed infrastructure and the presence of wireless communication introduce unique challenges and vulnerabilities. Here are some key aspects of security in MANETs:

Authentication and Authorization: MANET nodes should be able to authenticate each other to ensure that only legitimate nodes participate in the network. Authorization mechanisms control the access rights and privileges of different nodes.

Confidentiality: Encryption techniques are used to protect the confidentiality of data transmitted over the network. Encryption prevents unauthorized nodes from eavesdropping on the communication.

Data Integrity: Ensuring data integrity is crucial to prevent unauthorized modification of data during transmission. Techniques like digital signatures and message authentication codes help verify the integrity of data.

Availability: Attacks that aim to disrupt the availability of the network or its resources can be detrimental. Counter measures against denial-of-service (DoS) attacks and other forms of resource exhaustion are essential.

Routing Security: Since routing is dynamic in MANETs, securing the routing process is crucial. Attacks like route discovery flooding and black hole attacks can disrupt network communication. Secure routing protocols help mitigate such threats.

Intrusion Detection and Prevention: MANETs need mechanisms to detect and respond to intrusion attempts. Intrusion detection systems can identify suspicious behavior, while intrusion prevention systems take action to mitigate threats.

Key Management: Secure key distribution and management are essential for encryption and authentication. Distributing and updating keys in a dynamic network is challenging and requires efficient protocols.

Trust Establishment: Building trust among nodes is important to prevent attacks from malicious nodes. Reputation-based systems and trust models help nodes make informed decisions about whom to trust for communication and collaboration.

Physical Layer Security: The physical layer of wireless communication can be vulnerable to eavesdropping and interference. Techniques like physical-layer encryption and spread-spectrum modulation can enhance security.

Energy-Aware Security: In resource-constrained MANETs, security mechanisms should also consider energy consumption. Energy-efficient cryptographic algorithms and optimized security protocols are important.

Collaborative Security: Nodes can work together to enhance security by sharing information about detected threats or compromised nodes. Collaborative security mechanisms improve the overall network's ability to respond to attacks.

Network Partitioning: In the case of network partitioning due to node mobility or other factors, security measures should be in place to prevent unauthorized nodes from exploiting the situation.

Security in MANETs is a complex area that requires a combination of cryptographic techniques, secure protocols, intrusion detection mechanisms, and adaptive strategies. As MANETs continue to evolve, so do the challenges and solutions related to securing these dynamic and decentralized networks.

Check Your Progress 4

1. Authorization mechanisms control the access ----- of different nodes.
2. Collaborative security mechanisms improve the overall ----- to respond to attacks.
3. What is Data Integrity?

9.6 SUMMARY

Mobile Ad Hoc Networks (MANETs) are decentralized wireless networks formed by mobile devices without relying on fixed infrastructure. Their key objectives include dynamic communication, decentralization, flexibility, and robustness. MANETs are adaptable and scalable, making them valuable for emergency response, military operations, sensor networks, collaborative applications, and reducing infrastructure costs. These networks face challenges in routing, security, QoS, and energy efficiency due to their dynamic and resource-constrained nature. MANETs play a pivotal role in

9.7 SOLUTIONS / ANSWERS

Check Your Progress 1

1. MANETs allow for secure and dynamic communication among soldiers, vehicles, and other assets.
2. MANETs support collaborative applications such as sharing resources, data, and information among devices in close proximity.
3. What are the Applications of MANETs?
Decentralization, Dynamic Communication, flexibility & scalability, robustness & resilience, emergency & remote communication, military applications, sensor networks.

Check Your Progress 2

1. The goal of routing is to ensure that data packets reach their intended destination reliably and in a timely manner.
2. Route Discovery phase determines the most optimal path for the transmission of data packets between the source and the destination mobile nodes.
3. How to avoid vulnerable nodes?
Avoiding Vulnerable Nodes: Paths that involve nodes with known vulnerabilities or compromised security are avoided.

Check Your Progress 3

1. Congestion-aware algorithms detect and alleviate congestion to prevent network performance degradation. These mechanisms might involve rate limiting, flow control, and dynamic route adjustments.
2. Performance Metrics in QoS can be measured using throughput, delay, packet loss, jitter, and reliability.
3. What are the challenges in MANETs?
MANETs face challenges such as limited bandwidth, node mobility, energy constraints, and network partitioning. These challenges can impact the ability to maintain consistent QoS levels.

Check Your Progress 4

1. Authorization mechanisms control the access rights and privileges of different nodes.
2. Collaborative security mechanisms improve the overall network's ability to respond to attacks.
3. What is Data Integrity?

Ensuring data integrity is crucial to prevent unauthorized modification of data during transmission. Techniques like digital signatures and message authentication codes help verify the integrity of data.

9.8 FURTHER READINGS

1. "Ad Hoc Mobile Wireless Networks: Protocols and Systems" by C. K. Toh
2. "Mobile Ad Hoc Networking" by Stefano Basagni
3. "Ad Hoc Networks: Technologies and Protocols" by Prasant Mohapatra and Srikanth Krishnamurthy
4. "Mobile Ad Hoc Networking and Applications" edited by Karmakar, G., & Dooley, L. S.
5. "Wireless Communications and Mobile Computing" by Karmakar, G., & Zeadally, S.

****Websites and Online Resources:****

1. IEEE Xplore: The IEEE's digital library has a wealth of research papers, articles, and conference proceedings related to MANETs.

(Website: <https://ieeexplore.ieee.org/>)

2. ACM Digital Library: Similar to IEEE Xplore, the ACM Digital Library offers research papers and publications on various aspects of mobile ad hoc networks.

(Website: <https://dl.acm.org/>)

3. SpringerLink: Springer provides access to numerous books, journals, and articles on networking topics, including mobile ad hoc networks.

(Website: <https://www.springer.com/gp>)

4. MANET-related Conferences: Keep an eye on conferences like ACM MobiHoc, IEEE MASS, and IEEE WoWMoM, which focus on mobile ad hoc networks and wireless communications.

5. ResearchGate: You can find research papers, articles, and discussions related to MANETs on ResearchGate, a platform for researchers to collaborate and share findings.

(Website: <https://www.researchgate.net/>)

6. Coursera and edX: Look for online courses related to networking, wireless communication, or mobile ad hoc networks on platforms like Coursera and edX.

Remember that the field of mobile ad hoc networks is continuously evolving, so staying up-to-date with the latest research, publications, and advancements is essential. Always verify the credibility of the sources you use for your research and learning.